

SANTA MARIA AP, PortugalWMO: **085150**Lat: **36.973N**Lon: **25.173W**Elev: **97**StdP: **100.17**Time Zone: **-1.00 (AZR)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.7	11.2	4.8	5.4	12.6	5.8	5.8	13.1	16.0	15.2	14.5	14.8	5.2	60	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	26.2	21.9	25.8	21.7	25.0	21.2	23.2	25.0	22.7	24.5	22.2	24.1	5.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.4	24.2	22.0	16.9	23.9	21.4	16.3	23.4	69.1	25.2	67.1	24.4	65.4	24.0	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.2	10.8	DB	8.8	27.4	0.9	0.7	8.1	27.9	7.6	28.3	7.1	28.6	6.5	29.1	(4)
(5)			WB	6.8	24.2	0.8	0.9	6.2	24.9	5.7	25.4	5.2	25.9	4.7	26.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	15.2	14.8	15.1	15.9	17.3	19.5	21.7	22.9	22.1	20.1	17.6	16.1
	DBStd	3.16	1.51	1.39	1.37	1.29	1.39	1.36	1.33	1.15	1.28	1.62	1.74	1.58
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	522	97	99	100	74	38	5	0	0	0	4	35	69
	CDD10.0	2997	162	134	158	176	227	284	362	400	363	313	227	190
	CDD18.3	476	0	0	0	0	7	39	104	142	113	58	12	2
	CDH23.3	780	0	0	0	0	0	10	154	389	202	25	1	0
	CDH26.7	9	0	0	0	0	0	0	2	7	1	0	0	0
Wind	WSAvg	5.2	6.0	5.9	6.1	5.7	5.0	4.4	3.9	4.0	4.3	5.6	5.6	6.3
Precipitation	PrecAvg	740	90	73	76	54	32	26	20	33	54	94	89	98
	PrecMax	1302	286	223	261	140	133	88	113	161	160	260	335	292
	PrecMin	445	3	3	0	4	0	0	1	4	5	18	8	15
	PrecStd	164	62	52	57	41	29	22	21	31	33	61	64	59
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	19.0	19.2	20.4	22.2	24.2	26.8	27.2	26.6	25.0	23.0	20.5
		MCWB	17.0	16.4	15.6	16.6	18.5	20.0	22.0	22.4	22.2	21.5	20.0	18.0
	2%	DB	18.2	18.1	18.2	19.2	21.1	23.2	25.7	26.6	25.9	24.0	21.8	19.7
		MCWB	16.4	15.8	15.4	16.1	17.9	19.4	21.6	22.1	21.9	20.7	19.4	17.7
	5%	DB	18.0	17.4	17.9	18.8	20.4	22.8	24.9	26.0	25.1	23.2	20.9	19.0
		MCWB	16.2	15.4	15.2	15.9	17.3	19.2	21.1	21.8	21.4	20.2	18.7	17.4
	10%	DB	17.2	17.0	17.2	18.1	19.9	22.0	24.1	25.2	24.5	22.7	20.1	18.4
		MCWB	15.7	15.1	14.8	15.3	16.9	18.7	20.4	21.3	21.0	20.0	18.1	16.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	17.6	17.5	17.9	19.8	21.6	23.2	24.1	23.7	22.7	20.9	19.0
		MCDB	18.4	18.2	18.2	19.1	21.1	23.2	25.1	25.8	25.3	23.9	22.0	19.8
	2%	WB	17.3	16.7	16.7	17.3	19.0	20.7	22.6	23.4	23.1	21.8	20.1	18.4
		MCDB	18.0	17.5	17.6	18.6	20.3	22.2	24.6	25.2	24.6	23.0	21.1	19.2
	5%	WB	16.7	16.3	16.3	16.7	18.4	20.2	22.0	22.7	22.5	21.2	19.5	17.7
		MCDB	17.5	17.1	17.2	18.0	19.8	21.8	23.9	24.7	24.2	22.6	20.5	18.6
	10%	WB	16.4	15.6	15.6	16.3	17.7	19.5	21.4	22.3	21.9	20.6	18.7	17.4
		MCDB	17.3	16.6	16.8	17.6	19.2	21.2	23.4	24.4	23.7	22.0	19.8	18.3
Mean Daily Temperature Range	5% DB	MDBR	3.6	4.0	4.2	4.4	4.6	4.8	5.0	5.1	5.0	4.3	4.0	3.6
		MCDBR	3.5	4.2	4.6	4.8	4.9	5.1	5.4	5.4	5.3	4.7	4.3	3.8
		MCWBR	3.0	3.3	3.4	3.3	3.3	3.1	3.0	3.1	3.2	3.3	3.2	3.3
	5% WB	MCDBR	3.1	3.7	3.8	4.2	4.2	4.4	4.7	4.7	4.6	4.1	3.6	3.4
		MCWBR	3.2	3.6	3.5	3.3	3.4	3.4	3.3	3.4	3.6	3.5	3.4	3.5
Clear-Sky Solar Irradiance	taub	0.345	0.351	0.372	0.374	0.368	0.369	0.363	0.361	0.359	0.368	0.350	0.351	
	taud	2.547	2.530	2.466	2.470	2.514	2.548	2.571	2.599	2.610	2.563	2.557	2.524	
	Ebn at Noon	845	885	893	908	914	909	912	909	897	855	831	809	
	Edh at Noon	78	89	104	109	106	102	99	94	89	85	77	76	
All-Sky Solar Radiation	RadAvg	2.26	3.09	4.23	5.39	6.23	6.60	6.59	6.23	5.10	3.57	2.51	1.93	
	RadStd	0.18	0.19	0.28	0.35	0.38	0.39	0.35	0.25	0.22	0.20	0.16	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	+0.30	+0.44	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A